LA Lab - 5

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Aim : Logical Volume Management (LVM): Create LVM volumes, snapshots, and extend file systems without downtime.

Theory :

What is LVM?

Logical Volume Management (LVM) is a storage management technique in Linux that provides an abstract layer between physical storage devices and the file system. It allows flexible allocation, resizing, and management of disk space compared to traditional partitioning.

Why LVM is Used

Traditional disk partitioning has limitations such as fixed size and difficulty in resizing. LVM overcomes these issues by allowing:

- Dynamic resizing of partitions
- Combining multiple disks into one
- Creating backups using snapshots

- Managing storage without downtime

Architecture of LVM

LVM works in a three-layer structure:

1. Physical Volume (PV)

- The lowest layer in LVM
- Represents physical storage devices (hard disks or partitions)
- Example: `/dev/sda1`

2. Volume Group (VG)

- A pool of storage created by combining one or more PVs
- Acts like a virtual disk

3. Logical Volume (LV)

- Created from a Volume Group
- Works like a normal partition
- Can be formatted and mounted

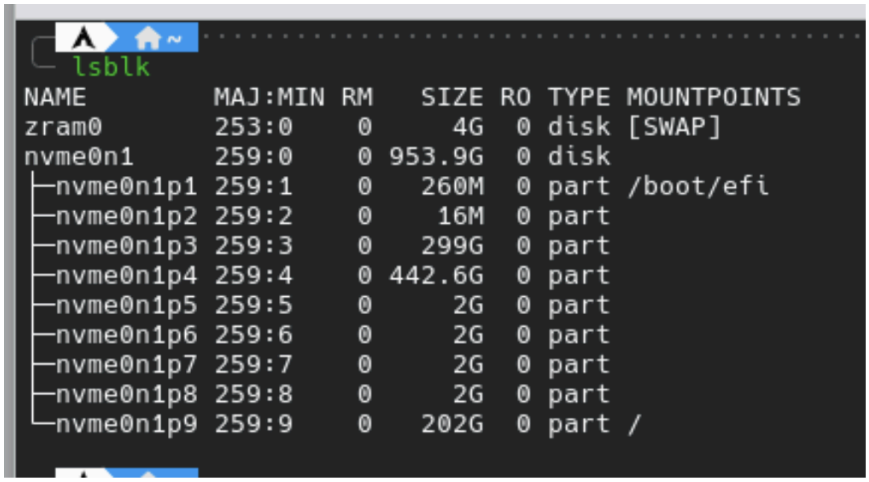
Flow:
PV → VG → LV → File System → Mount Point

Task 1: Explore Storage Devices

Before working with LVM, it is important to inspect available storage devices and ensure that the target partition is not mounted. Mounted partitions cannot be modified for LVM operations.

In this task, we listed all available block devices using `lsblk` and verified the target partition. The partition /dev/nvme0n1p10was unmounted using the umount command to prepare it for LVM configuration.

lsblk
sudo umount /dev/nvme0n1p10

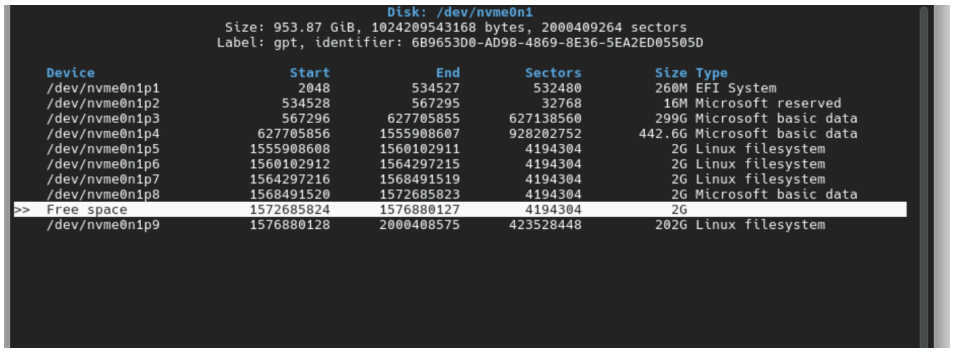


```
lsblk
NAME        MAJ:MIN RM   SIZE RO TYPE MOUNTPOINTS
zram0       253:0    0     4G  0 disk [SWAP]
nvme0n1     259:0    0 953.9G  0 disk
├nvme0n1p1 259:1    0   260M  0 part /boot/efi
├nvme0n1p2 259:2    0    16M  0 part
├nvme0n1p3 259:3    0   299G  0 part
├nvme0n1p4 259:4    0 442.6G  0 part
├nvme0n1p5 259:5    0     2G  0 part
├nvme0n1p6 259:6    0     2G  0 part
├nvme0n1p7 259:7    0     2G  0 part
├nvme0n1p8 259:8    0     2G  0 part
└nvme0n1p9 259:9    0   202G  0 part /
```

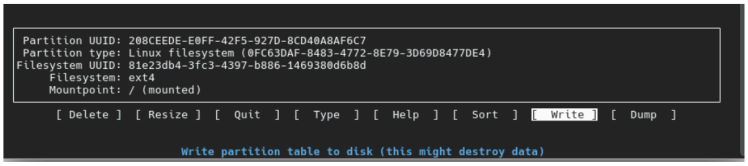
Task 2: Create Disk Partition for LVM

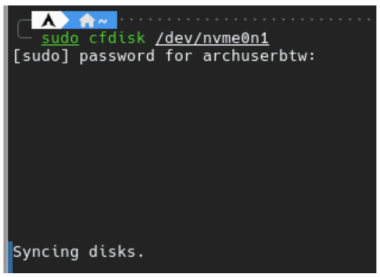
To use a disk in LVM, it must first be initialized as a Physical Volume (PV). This process labels the partition so it can be used by LVM.
We initialized the partition /dev/nvme0n1p10 as a Physical Volume using pvcreate, making it ready for inclusion in a Volume Group.

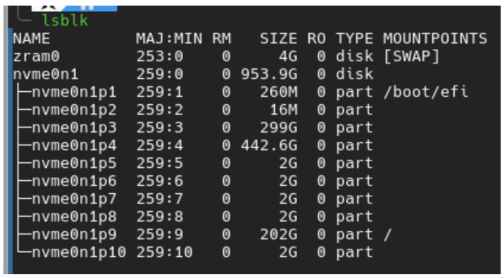
sudo pvcreate /dev/nvme0n1p10



```
                                Disk: /dev/nvme0n1
                 Size: 953.87 GiB, 1024209543168 bytes, 2000409264 sectors
                Label: gpt, identifier: 6B9653D0-AD98-4869-8E36-5EA2ED05505D

   Device                     Start          End      Sectors       Size Type
   /dev/nvme0n1p1              2048       534527       532480       260M EFI System
   /dev/nvme0n1p2            534528       567295        32768        16M Microsoft reserved
   /dev/nvme0n1p3            567296    627705855    627138560       299G Microsoft basic data
   /dev/nvme0n1p4         627705856   1555908607    928202752     442.6G Microsoft basic data
   /dev/nvme0n1p5        1555908608   1560102911      4194304        2G Linux filesystem
   /dev/nvme0n1p6        1560102912   1564297215      4194304        2G Linux filesystem
   /dev/nvme0n1p7        1564297216   1568491519      4194304        2G Linux filesystem
   /dev/nvme0n1p8        1568491520   1572685823      4194304        2G Microsoft basic data
>> Free space           1572685824   1576880127      4194304        2G
   /dev/nvme0n1p9        1576880128   2000408575    423528448       202G Linux filesystem
```







Task 3: Create Physical Volume (PV)

A Physical Volume is the basic storage unit in LVM. It represents a disk or partition that is brought under LVM control.

The partition was successfully converted into a Physical Volume, forming the base layer of the LVM structure.

sudo pvcreate /dev/nvme0n1p10

Task 4: Create Volume Group (VG) & Logical Volume (LV)

- Volume Group (VG): A pool of storage created from one or more PVs.
- Logical Volume (LV): A virtual partition created from the VG.

We created a Volume Group named vg_test using the Physical Volume. Then, a Logical Volume lv_data of size 512MB was created from this group.

sudo vgcreate vg_test /dev/nvme0n1p10

sudo lvcreate -L 512M -n lv_data vg_test

```
sudo pvcreate /dev/nvme0n1p10
Physical volume "/dev/nvme0n1p10" successfully created.
```

```
sudo vgcreate vg_test /dev/nvme0n1p10
Volume group "vg_test" successfully created
```

```
sudo lvcreate -L 256M -n lv_data vg_test
Logical volume "lv_data" created.
```

Task 5: Format and Mount LV

Before using a Logical Volume, it must be formatted with a file system and mounted to a directory.

The Logical Volume was formatted using the ext4 file system. A mount point /mnt/lvm_test was created, and the LV was mounted successfully for usage.

sudo mkfs.ext4 /dev/vg_test/lv_data

sudo mkdir /mnt/lvm_test

sudo mount /dev/vg_test/lv_data /mnt/lvm_test

```
sudo mkfs.ext4 /dev/vg_test/lv_data
mke2fs 1.47.4 (6-Mar-2025)
Discarding device blocks: done
Creating filesystem with 262144 1k blocks and 65536 inodes
Filesystem UUID: b410a830-b499-4d03-ac6c-4deddfcda3fa
Superblock backups stored on blocks:
        8193, 24577, 40961, 57345, 73729, 204801, 221185

Allocating group tables: done
Writing inode tables: done
Creating journal (8192 blocks): done
Writing superblocks and filesystem accounting information: done
```

```
sudo mkdir /mnt/lvm_test

sudo mount /dev/vg_test/lv_data /mnt/lvm_test

lsblk
NAME                MAJ:MIN RM   SIZE RO TYPE MOUNTPOINTS
zram0               253:0    0     4G  0 disk [SWAP]
nvme0n1             259:0    0 953.9G  0 disk
├─nvme0n1p1         259:1    0   260M  0 part /boot/efi
├─nvme0n1p2         259:2    0    16M  0 part
├─nvme0n1p3         259:3    0   299G  0 part
├─nvme0n1p4         259:4    0 442.6G  0 part
├─nvme0n1p5         259:5    0     2G  0 part
├─nvme0n1p6         259:6    0     2G  0 part
├─nvme0n1p7         259:7    0     2G  0 part
├─nvme0n1p8         259:8    0     2G  0 part
├─nvme0n1p9         259:9    0   202G  0 part /
└─nvme0n1p10        259:10   0     2G  0 part
  └─vg_test-lv_data 252:0    0   256M  0 lvm  /mnt/lvm_test
```

Task 6: Extend Logical Volume (Online)

LVM allows dynamic resizing of Logical Volumes without unmounting them. This is known as online resizing.

We extended the Logical Volume by 256MB using lvextend with the --resizefs option, which also resized the file system automatically without downtime.

sudo lvextend -L +256M /dev/vg_test/lv_data --resizefs

```
sudo lvextend -L +256M /dev/vg_test/lv_data --resizefs
  File system ext4 found on vg_test/lv_data mounted at /mnt/lvm_test.
  Size of logical volume vg_test/lv_data changed from 256.00 MiB (64 extents) to 512.00 MiB (128 extents).
  Extending file system ext4 to 512.00 MiB (536870912 bytes) on vg_test/lv_data...
resize2fs /dev/vg_test/lv_data
resize2fs 1.47.4 (6-Mar-2025)
Filesystem at /dev/vg_test/lv_data is mounted on /mnt/lvm_test; on-line resizing required
old_desc_blocks = 2, new_desc_blocks = 4
The filesystem on /dev/vg_test/lv_data is now 524288 (1k) blocks long.

resize2fs done
  Extended file system ext4 on vg_test/lv_data.
  Logical volume vg_test/lv_data successfully resized.
```

Task 7: Add New Disk to Volume Group

Additional storage can be added to a Volume Group by initializing a new PV and extending the VG.
A new disk partition was initialized as a PV and added to the existing Volume Group using vgextend, increasing the available storage pool.

sudo pvcreate /dev/nvme0n1p5

sudo vgextend vg_test /dev/nvme0n1p5

```
sudo pvcreate /dev/nvme0n1p5
Physical volume "/dev/nvme0n1p5" successfully created.

sudo vgextend vg_test /dev/nvme0n1p5
Volume group "vg_test" successfully extended
```

Task 8: Create Multiple Logical Volumes

Multiple Logical Volumes can be created within a single Volume Group to organize storage efficiently.

We created an additional Logical Volume lv_backup from the same Volume Group for backup purposes.

sudo lvcreate -L 512M -n lv_backup vg_test

Task 9: Display LVM Information

LVM provides commands to display details about PVs, VGs, and LVs.

We used vgs and lvs commands to display the structure and status of Volume Groups and Logical Volumes.

sudo vgs

sudo lvs

Task 10: Rename VG / LV

LVM allows renaming of Volume Groups and Logical Volumes without affecting stored data.

We renamed the Logical Volume and Volume Group using lvrename and vgrename commands successfully.

sudo lvrename vg_test lv_data lv_production

sudo vgrename vg_test vg_real

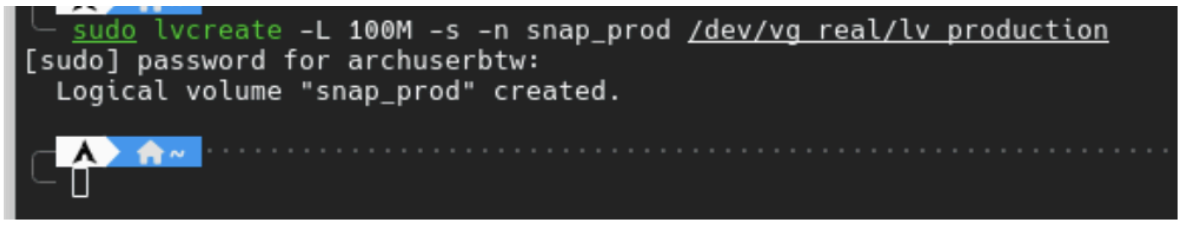
```
sudo lvcreate -L 512M -n lv_backup vg_test
Logical volume "lv_backup" created.

sudo vgs
VG      #PV #LV #SN Attr    VSize VFree
vg_test   2   2   0 wz--n-- 3.99g 2.99g

sudo lvs
LV        VG      Attr       LSize   Pool Origin Data%  Meta%  Move Log Cpy%Sync Convert
lv_backup vg_test -wi-a----- 512.00m
lv_data   vg_test -wi-ao---- 512.00m

sudo lvrename vg_test lv_data lv_production
Renamed "lv_data" to "lv_production" in volume group "vg_test"

sudo lvs
LV            VG      Attr       LSize   Pool Origin Data%  Meta%  Move Log Cpy%Sync Convert
lv_backup     vg_test -wi-a----- 512.00m
lv_production vg_test -wi-ao---- 512.00m
```

```
sudo lvrename vg_test lv_data lv_production
Renamed "lv_data" to "lv_production" in volume group "vg_test"

sudo lvs
LV            VG      Attr       LSize   Pool Origin Data%  Meta%  Move Log Cpy%Sync Convert
lv_backup     vg_test -wi-a----- 512.00m
lv_production vg_test -wi-ao---- 512.00m

sudo vgrename vg_test vg_real
Volume group "vg_test" successfully renamed to "vg_real"

sudo vgs
VG      #PV #LV #SN Attr    VSize VFree
vg_real   2   2   0 wz--n-- 3.99g 2.99g
```

Task 11: Create LVM Snapshot

Snapshots are point-in-time copies of Logical Volumes using Copy-on-Write (CoW) technology.
A snapshot snap_prod was created for the Logical Volume to preserve its current state before performing risky operations.

sudo lvcreate -L 100M -s -n snap_prod /dev/vg_real/lv_production



Task 12: Restore Snapshot

Snapshots can be merged back to restore the original state of the Logical Volume.

We unmounted the volume and used lvconvert --merge to restore data from the snapshot, effectively rolling back changes.

sudo umount /mnt/lvm_test

sudo lvconvert --merge /dev/vg_real/snap_prod

sudo mount /dev/vg_real/lv_production /mnt/lvm_test

```
sudo umount /mnt/lvm_test

sudo lvconvert --merge /dev/vg_real/snap_prod
Merging of volume vg_real/snap_prod started.
WARNING: Failed to connect to lvmpolld. Proceeding with polling without using lvmpolld.
WARNING: Check global/use_lvmpolld in lvm.conf or the lvmpolld daemon state.
vg_real/lv_production: Merged: 100.00%
Merge of snapshot into logical volume vg_real/lv_production has finished.
Logical volume "snap_prod" successfully removed.

sudo mount /dev/vg_real/lv_production /mnt/lvm_test
```

Task 13: Reduce Logical Volume

Reducing a Logical Volume requires shrinking the file system first, followed by reducing the LV size.

We performed file system check, reduced its size, and then decreased the Logical Volume size safely.

sudo umount /mnt/lvm_test

sudo e2fsck -f /dev/vg_real/lv_production

sudo resize2fs /dev/vg_real/lv_production 128M

sudo lvreduce -L 128M /dev/vg_real/lv_production

```
└─ sudo e2fsck -f /dev/vg_real/lv_production
e2fsck 1.47.4 (6-Mar-2025)
Pass 1: Checking inodes, blocks, and sizes
Pass 2: Checking directory structure
Pass 3: Checking directory connectivity
Pass 4: Checking reference counts
Pass 5: Checking group summary information
/dev/vg_real/lv_production: 12/131072 files (8.3% non-contiguous), 43690/524288 blocks

└─ sudo resize2fs /dev/vg_real/lv_production 128M
resize2fs 1.47.4 (6-Mar-2025)
Resizing the filesystem on /dev/vg_real/lv_production to 131072 (1k) blocks.
The filesystem on /dev/vg_real/lv_production is now 131072 (1k) blocks long.

└─ sudo lvreduce -L 128M /dev/vg_real/lv_production
  File system ext4 found on vg_real/lv_production.
  File system size (128.00 MiB) is equal to the requested size (128.00 MiB).
  File system reduce is not needed, skipping.
  Size of logical volume vg_real/lv_production changed from 512.00 MiB (128 extents) to 128.00 MiB (32 extents).
  Logical volume vg_real/lv_production successfully resized.
```

Task 14: Remove Logical Volume

Logical Volumes can be deleted when no longer needed.
We removed the Logical Volume using lvremove, freeing space in the Volume Group.

sudo lvremove /dev/vg_real/lv_production

Task 15: Remove Volume Group

Volume Groups can be deleted after removing all Logical Volumes.

The Volume Group was successfully removed using vgremove.

sudo vgremove vg_real

Task 16: Remove Physical Volume

Physical Volumes can be cleared of LVM metadata and returned to normal partitions.

```
sudo lvremove /dev/vg_real/lv_production
Do you really want to remove active logical volume vg_real/lv_production? [y/n]: y
  Logical volume "lv_production" successfully removed.

sudo lvs
  LV          VG      Attr       LSize   Pool Origin Data%  Meta%  Move Log Cpy%Sync Convert
  lv_backup   vg_real -wi-a----- 512.00m
  lv_mirrored vg_real rwi-a-r---  32.00m                                    100.00
  lv_striped  vg_real -wi-a-----  32.00m

sudo vgremove vg_real
Do you really want to remove volume group "vg_real" containing 3 logical volumes? [y/n]: y
Do you really want to remove active logical volume vg_real/lv_backup? [y/n]: y
  Logical volume "lv_backup" successfully removed.
Do you really want to remove active logical volume vg_real/lv_striped? [y/n]: y
  Logical volume "lv_striped" successfully removed.
Do you really want to remove active logical volume vg_real/lv_mirrored? [y/n]: y
  Logical volume "lv_mirrored" successfully removed.
  Volume group "vg_real" successfully removed
```

We removed LVM configuration from the disks using pvremove.

sudo pvremove /dev/nvme0n1p10

sudo pvremove /dev/nvme0n1p5

```
sudo vgs

sudo lvs
```

```
sudo pvremove /dev/nvme0n1p10
  Labels on physical volume "/dev/nvme0n1p10" successfully wiped.

sudo pvremove /dev/nvme0n1p5
  Labels on physical volume "/dev/nvme0n1p5" successfully wiped.
```

Task 17: Striping (RAID 0)

Striping distributes data across multiple disks to improve performance.
A striped Logical Volume was created using two disks, enabling parallel data writing for better speed.

sudo lvcreate -i 2 -L 32M -n lv_striped vg_real

Task 18: Mirroring (RAID 1)

Mirroring duplicates data across disks to provide redundancy and fault tolerance.

A mirrored Logical Volume was created, ensuring real-time backup of data across two disks.

sudo lvcreate -m 1 -L 32M -n lv_mirrored vg_real

```
sudo lvcreate -i 2 -L 32M -n lv_striped vg_real
Using default stripesize 64.00 KiB.
Logical volume "lv_striped" created.

sudo lvcreate -m 1 -L 32M -n lv_mirrored vg_real
Logical volume "lv_mirrored" created.

sudo lvs
LV            VG      Attr       LSize   Pool Origin Data%  Meta%  Move Log Cpy%Sync Convert
lv_backup     vg_real -wi-a----- 512.00m
lv_mirrored   vg_real rwi-a-r---  32.00m                                    100.00
lv_production vg_real -wi-a----- 128.00m
lv_striped    vg_real -wi-a-----  32.00m
```

Task 19: Automount LVM at Boot

To make a file system mount automatically at boot, its entry must be added to /etc/fstab using its UUID.

We retrieved the UUID using blkid and added an entry in /etc/fstab to enable automatic mounting of the Logical Volume at system startup.

blkid /dev/vg_real/lv_production

UUID=uuid_id/mnt/lvm_test ext4 defaults 0 2

```
sudo blkid /dev/vg_test/lv_data
/dev/vg_test/lv_data: UUID="ea929e20-5362-458e-b698-ef80998eb46b" BLOCK_SIZE="1024" TYPE="ext4"

sudo nano /etc/fstab
```

```
# /dev/vg_test/lv_data
UUID=ea929e20-5362-458e-b698-ef80998eb46b /mnt/lvm_test  ext4  defaults  0  2
```

<u>Conclusion</u> : This experiment demonstrated how Logical Volume Management (LVM) provides flexible and efficient disk management compared to traditional partitioning. We successfully created, extended, and managed logical volumes without system downtime. Features like snapshots, striping, and mirroring enhance data protection and performance.